Use of Remote Sensing in Crop Forecasting and Assessment of Impact of Natural Disasters: Operational Approaches in India

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Abstract:

Use of remote sensing data for pre-harvest crop production forecast has been operationalized in India under a national level programme called Forecasting Agricultural output using Space, Agrometeorology and Land based observations (FASAL). FASAL programme, funded by the Ministry of Agriculture, envisages multiple production forecasts of major crops of the country at National/State/District level. Similarly, under National Agricultural Drought Assessment and Monitoring System (NADAMS) project remote sensing data from multiple sources integrated with ground and meteorological information for district/sub-district level drought assessment for 13 states of India. The methodology for both of these programmes were developed by the Indian Space Research Organization and later on transferred to a centre (Mahalanobis National Crop Forecast Centre) specifically created under the Ministry of Agriculture to operationalize space technology applications in agriculture.

Keywords: FASAL, NADAMS, Crop Forecasting, Drought Assessment, Remote Sensing

1. Introduction

In India, agriculture is one of the major application areas of the remote sensing technology. Various national level agricultural applications have been developed which showcases the use of remote sensing data provided by the sensors/satellites launched by the country's space agency, Indian Space Research Organisation (ISRO). Some of these applications include, crop acreage and production estimation, cropping system analysis, agricultural water management, drought assessment and monitoring, horticultural development, precision farming, soil resources mapping, potential fishing zone forecast, watershed development, climate impact on agriculture and so on (Navalgund and Ray, 2000, Panigrahy and Ray, 2006, Navalgund et al., 2007). A few of these applications, after reaching operational level, have been transferred to the user departments. This has resulted in the institutionalization of the remote sensing applications in the country (Parihar & Manjunath, 2013). One recent example of operationalization/ institutionalization of remote sensing application is the creation of a centre called Mahalanobis National Crop Forecast Centre (MNCFC) under Department of Agriculture & Cooperation, Ministry of Agriculture. The centre operationalises two major programmes developed by ISRO on crop forecasting and drought assessment apart from various other activities related to agricultural assessment. This article describes these operational approaches.

2. Crop Forecasting

Crop forecasting is essential for various agricultural planning purposes, including pricing, export/import, contingency measures, etc. Crop forecasting using remote sensing data, in India, started in late 80s in Space Applications Centre of ISRO under the Department of Agriculture & Cooperation (DAC)'s sponsored project CAPE (Crop Acreage and Production Estimation). This later on developed into a national level programme, called FASAL (Forecasting Agriculture using Space, Agro-meteorology and Land based observations), which is in operation since August, 2006. FASAL project aims at providing multiple pre-harvest production forecasts of crops at National/State/District level (Parihar and Oza, 2006). Remote sensing data, both optical and microwave form the core of crop area enumeration, crop condition assessment and production forecasting. Crop yield is estimated using agrometeorological/spectral yield models and also crop growth simulation models. FASAL approach also involves using econometric models to forecast the area and production before the crop sowing operations. FASAL is a multi-institutional programme, which integrates the activity from many organizations. The list of such organizations is given in Table 1. MNCFC has started providing crop forecasts, under FASAL programme, from the Kharif (Rainy) season of 2012.

Table 1. Organizations involved in FASAL programme and their responsibilities

S.N.	Name of the Organisation	Responsibility
1	Directorate of Economics & Statistics, Ministry of Agriculture	Overall Coordination
2	Mahalanobis National Crop Forecast Centre, Ministry of Agriculture	Operational Crop Assessment
3	Space Applications Centre, Indian Space Research Organisation	Techniques Development
4	India Meteorological Department	Crop Yield Forecasting
5	Institute of Economic Growth	Econometric Modelling for pre-season forecast
6	State Agricultural Departments	Ground truth collection
7	State Remote Sensing Application Centres	Support for Techniques Development Activities

Currently, under FASAL project, National and State Level multiple forecasts are being issued for 5 crops (Rice (Kharif & Rabi), Jute, Rapeseed & Mustard, Winter Potato and Wheat). From the year 2013-14 onwards, State and District Level forecasts are generated for Cotton, Sugarcane and Rabi (winter season) Sorghum. While multi-date SAR (Synthetic Aperture Radar) data of Indian SAR satellite RISAT-1 is used for Rice (Kharif & Rabi) (Figure 2) and Jute, multi-date Resourcesat-2 AWiFS (Advanced Wide Field Sensor) data, with 56 m spatial resolution, are used for other crops. LISS III data, with 23.5 m resolution, is being used for district level assessments. Stratified random sampling approach is followed for crop area estimation. All those states, which together contribute more than 90% of the particular

crop's area in the country, are considered for area and production assessment. Hierarchical/logical classification approach is followed for classifying multi-date SAR data. A hybrid (combination of supervised and unsupervised) classification method is followed for multi-date optical data, while maximum likelihood (MXL) is followed for single date optical data. Indigenously developed software, called FASALSoft (Manthira Moorthi et al., 2014) is used for carrying out the digital analysis of remote sensing data. The yield forecasts are generated under FASAL programme by India Meteorological Department, correlation weighted empirical agro-meteorological models. For Rice and Wheat crops the final yield forecasts are being given using physical models with inputs from remote sensing data (Chakraborty et al., 2005; Tripathy et al., 2013). The details of the crop forecasts given in the year 2013-14 are presented in Table 2. Typically, in an agricultural year, 16 crop forecasts are generated. A comparison with the official estimates of Ministry of Agriculture showed that the Remote sensing based area estimates, at national level, are within -0.5 to -8.8 %, while differences in production estimates ranged between -11.2 to 5.6 %.

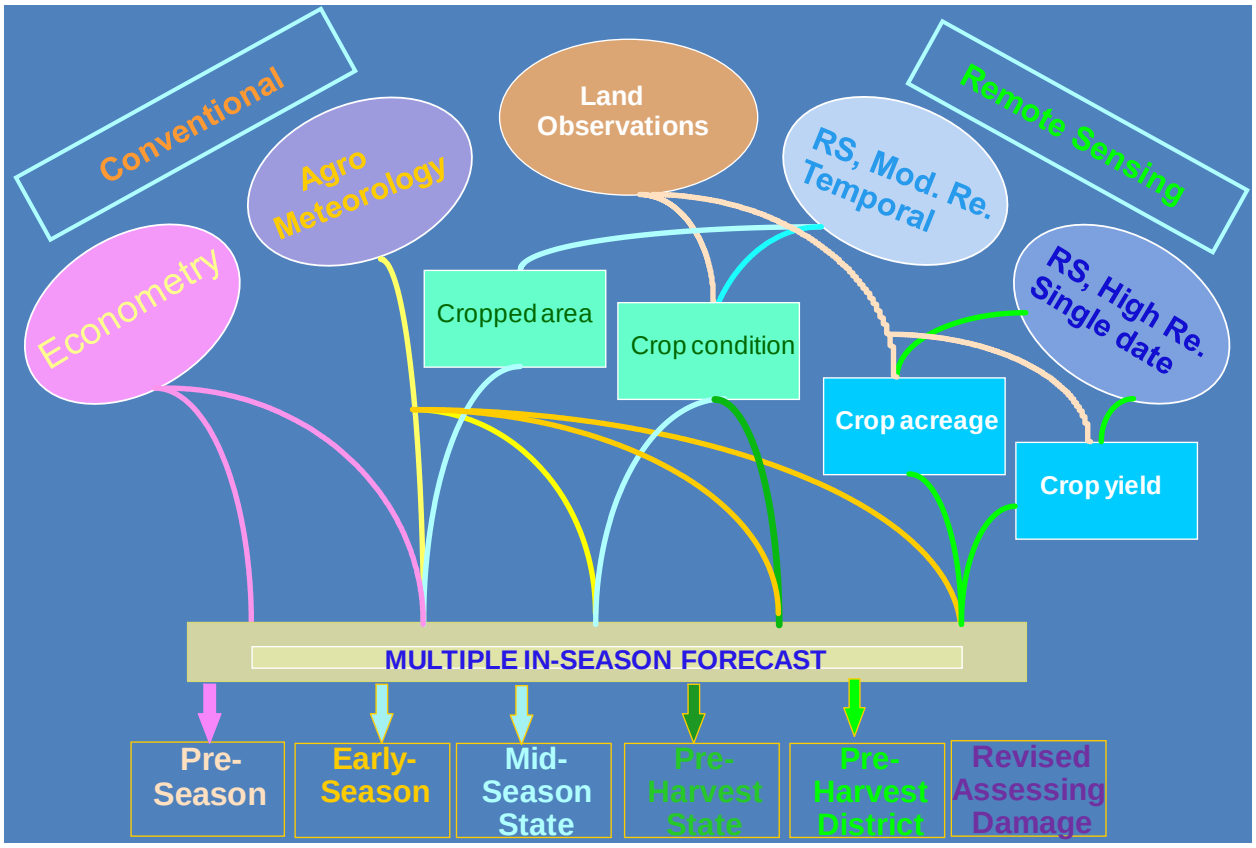


Figure 1. Approaches followed for multiple forecasts under FASAL programme (Source: SAC, ISRO)

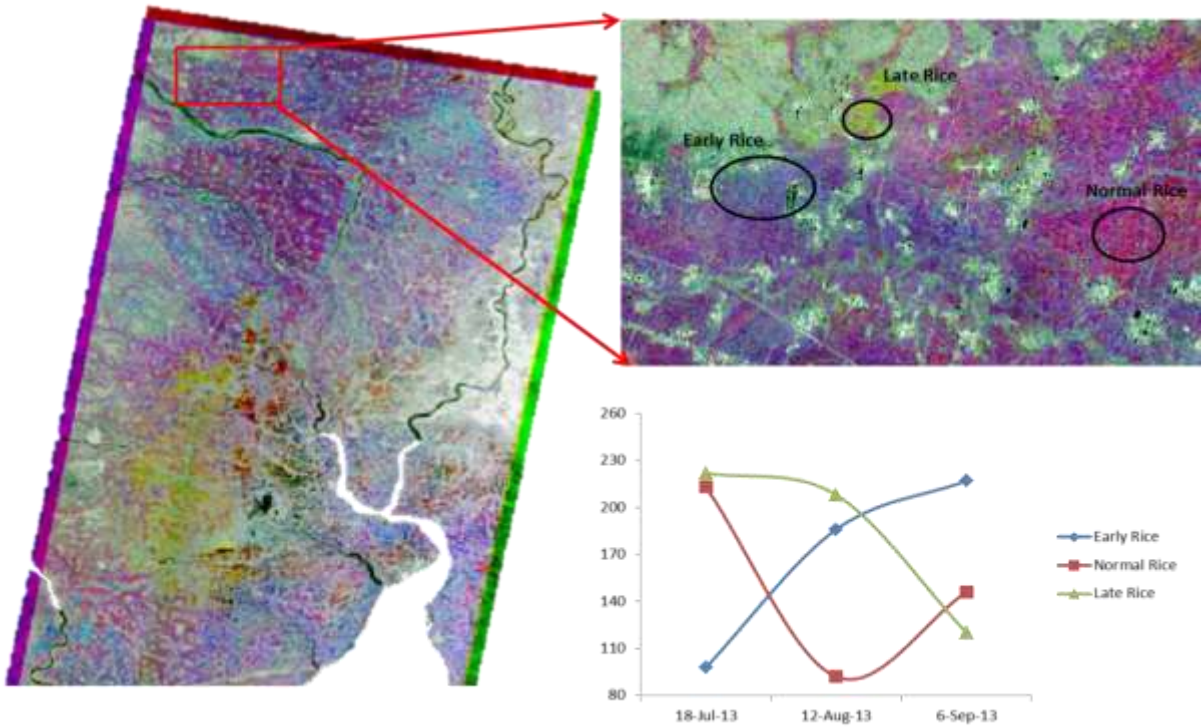


Figure 2. A typical 3-date RISAT-1 colour composite showing Rice crop of different transplanting time

Table 2. Summary of crop forecasts generated under FASAL project during 2013-14

Crop	No. of States	Forecasts	Dates of Forecasts	Satellite Data Used
Jute	3	F1	Jul 24, 2013	RISAT-1 SAR
Rice (Kharif)	13	F1 F2 F3	Aug 31, 2013 Sep 30, 2013 Jan 22, 2014	RISAT-1 SAR RISAT-1 SAR RISAT-1 SAR
Sugarcane	3	F1	Dec 18, 2013	R2 LISS III
Cotton	7	F1	Dec 18, 2013	R2 LISS III
Rapeseed & Mustard	5 5 6	F1 F2 F3	Dec 31, 2013 Jan 31, 2014 Feb 28, 2014	R2AWiFS R2 AWiFS R2 LISS III
Winter Potato	4 5	F1 F2	Jan 31, 2014 Mar 04, 2014	R2 AWiFS R2 LISS III
Wheat	9	F1 F2 F3	Feb 15, 2014 Mar 11, 2014 Apr 04, 2014	R2AWiFS R2 AWiFS R2 LISS III
Sorghum (Rabi)	2	F1	Feb 15, 2014	R2 LISS III
Rabi Rice	4	F1	Apr 04, 2014	RISAT-1 SAR

R2- Resourcesat-2 satellite

2.1 Ground truth collection using smart phone

Ground truth (GT) is an essential component for remote sensing data analysis. Ground truth includes collection of GPS Reading, Photographs, preparing a Sketch of the field Layout and filling-up of GT form. Earlier GT was being collected using a GPS, a camera and filling up of a form. A recent initiative was made for Smartphone based GT collection. An Android based application was developed by National Remote Sensing Centre of ISRO (Figure 3a). State Agriculture Department officials collected GT using Smartphones provided by MNCFC. The GT information directly comes to Bhuvan server (ISRO's Geoportal), which can be downloaded and used real-time (Figure 3b). During Kharif and Rabi season of 2013-14 more than 5800 GT points were collected from 16 states using the smart phones.

Figure 3. Real time ground truth collection using smart phone, a) Android App developed by NRSC, b) Ground Truth sites available on Bhuvan server

2.2 Remote sensing based Crop Cutting Experiments

Crop cutting experiments (CCE) are carried for each crop to estimate crop yield at district and state level. The conventional method CCE planning does not consider the current crop condition, which may result error in sampling. To overcome this, remote sensing driven Crop Cutting Experiments (CCE) planning was carried out in Bihar state for Rice crop during Kharif season of 2013. The work was carried out jointly with Bihar Agriculture Department. Rice Crop Map was generated using such as RISAT-1 MRS data. Resourcesat-2 AWiFS time composite NDVI during September 2nd Fortnight to October 1st Fortnight was extracted for rice area. Three classes (A, B, C) were defined based on frequency distribution of NDVI values. 37 points were randomly selected in 22 districts of Bihar (Figure 4a). Crop cutting experiment was carried out under the supervision of MNCFC. Yield models were developed between NDVI and yield and yield map was generated (Figure 4b). The study showed very high efficiency of stratified sampling plan generated for CCE based on NDVI values.

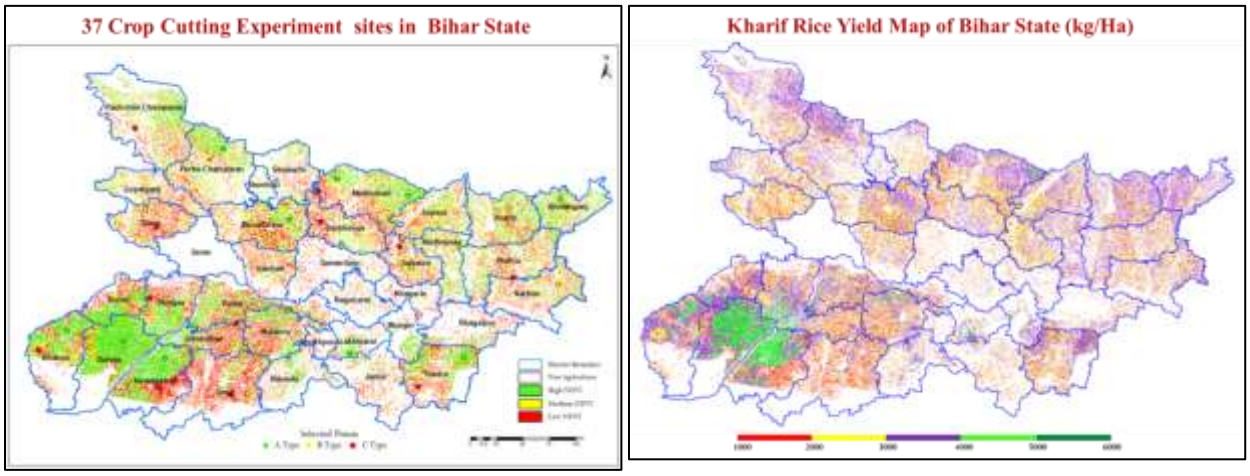


Figure 4. Use of remote sensing data for a) selecting sites for Crop Cutting Experiments for rice crop in Bihar and b) generating a yield map

2.3 Crop Emergence Progression

The NDVI (Normalized Difference Vegetation Index) product derived from the Indian geostationary satellite INSAT-3A based CCD camera is extremely useful for vegetation monitoring, because of its high temporal frequency (half an hour) (Nigam et al., 2011). The spatial resolution of the data is 1 km. The 10 day NDVI product was used to monitor the Rabi season crop emergence based on a methodology developed by Space Applications Centre, ISRO (Vyas et al., 2011). Since Satellite data sees the crop only after spectral emergence (i.e. the time when crops are big enough to start registering spectral signature) this was called as emergence area progression. The analysis was carried out for 6 states (Punjab, Haryana, Rajasthan, UP, MP and Bihar) at every 10 day interval during December, 2013 to February, 2014 (Figure 5).

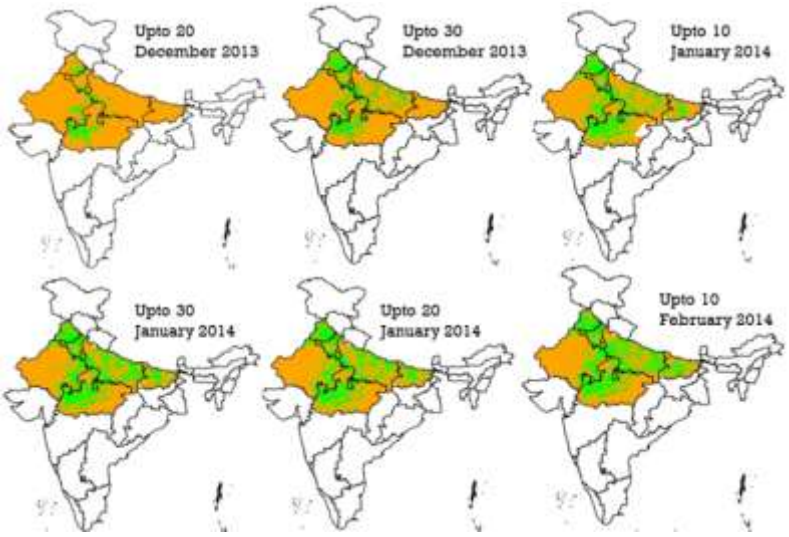


Figure 5. Map of Rabi season crop Emergence Area progression in 6 States of India

3. Drought and Flood Impact Assessment on Agriculture

Agriculture, in India, is strongly affected by two major hydro-meteorological disasters, namely drought and flood. Drought is a perennial feature. 16 per cent of India's total area is drought prone and approximately, 50 million people are annually affected by droughts (DAC, 2009). Over 68-70% of total sown area in India is vulnerable to drought. Similarly, around 40 million hectares of land in India is prone to floods as per National Flood Commission report.

Assessment of agricultural condition during drought or flood is essential for taking various relief and rehabilitation measures. Since both these disasters impact large area, satellite based monitoring is extremely useful.

3.1 Agricultural Drought Assessment

In India, operational agricultural drought assessment using remote sensing data is carried under a major programme called National Agricultural Drought Assessment & Monitoring System (NADAMS). NADAMS project, developed by National Remote Sensing Centre, provides near real-time information on prevalence, severity level and persistence of agricultural drought at state/ district/sub-district level (Murthy & Sesha Sai, 2011). Currently, it covers 13 states of India (Andhra Pradesh, Bihar, Chattisgarh, Gujarat, Haryana, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Odisha, Rajasthan, Tamil Nadu, and Uttar Pradesh), which are predominantly agriculture based and prone to drought situation. In four states (Andhra Pradesh, Karnataka, Haryana and Maharashtra), the assessment is carried out at sub-district level. The remote sensing data of NOAA AVHRR (for district level), MODIS and Resourcesat-2 Advanced Wide Field Sensor, AWiFS (for sub-district level) along with rainfall data are used for drought assessment (Choudhary et al., 2011). Various spectral indices, such as Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) & Shortwave Angle Slope Index (SASI) are computed and integrated with Soil Moisture Index and District Level Rainfall to assess the drought condition (Figure 6). Agricultural conditions are monitored at state/district level using daily NOAA AVHRR/ MODIS data. Fortnightly/monthly report of drought condition is provided to all the concerned central and state government agencies under NADAMS. MNCFC has started providing periodic Drought Assessment Reports from the Kharif season of 2012 (Ray et al., 2014).

The district level drought assessment map for September, 2013 is presented in figure 7, which shows in the 13 states 11 districts were in moderate drought condition and 43 districts in mild drought condition. Every month, similar maps are generated and circulated along with all satellite derived products. Recently a drought geo-portal has been created (www.ncfc.gov.in), where users can access the images, maps, assessments and NDVI profiles.

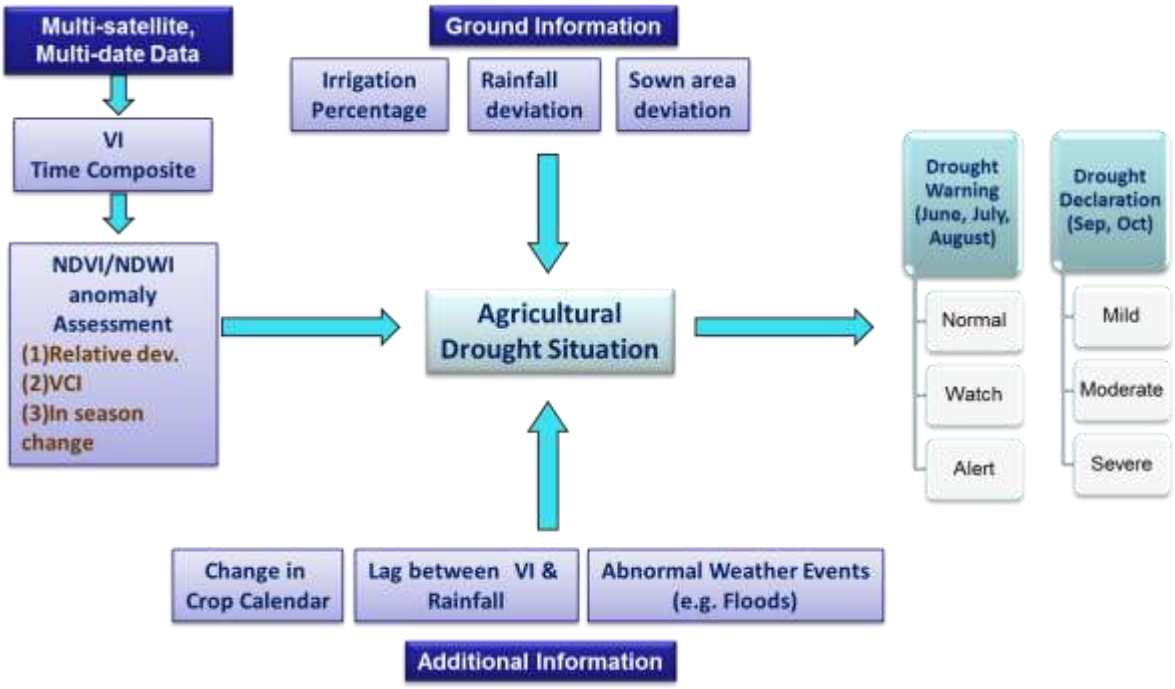
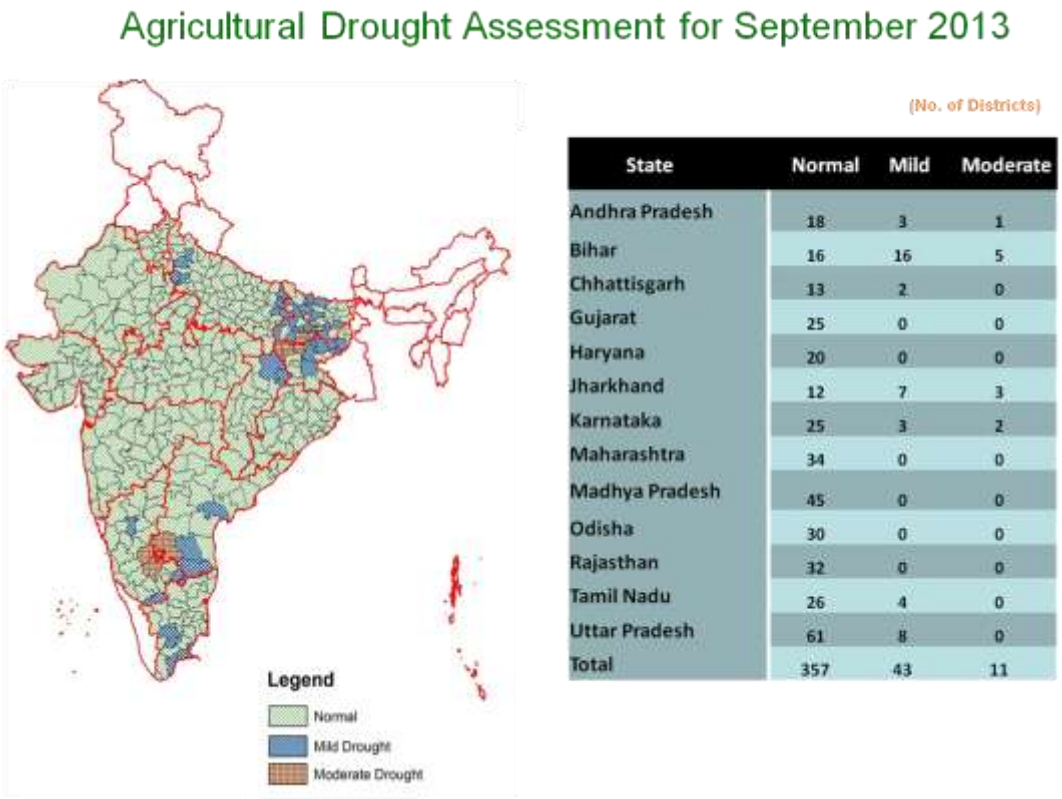


Figure 6. Methodology for Drought Assessment under NADAMS project



(No. of Districts)

State	Normal	Mild	Moderate
Andhra Pradesh	18	3	1
Bihar	16	16	5
Chhattisgarh	13	2	0
Gujarat	25	0	0
Haryana	20	0	0
Jharkhand	12	7	3
Karnataka	25	3	2
Maharashtra	34	0	0
Madhya Pradesh	45	0	0
Odisha	30	0	0
Rajasthan	32	0	0
Tamil Nadu	26	4	0
Uttar Pradesh	61	8	0
Total	357	43	11

Figure 7. District level drought assessment during September, 2013 under NADAMS project

3.2 Rice Flooded Area Mapping

Another study was carried out to assess the impact of flood on Rice crop in Odisha state of India post Phailin cyclone of October 12, 2013. RISAT SAR derived rice map was integrated with flood inundation map developed National Remote Sensing Centre to map the areas of rice crop under flood. Total ten districts were affected by cyclone and rice inundation was more than 4% of rice area. Four Districts namely Baleshwar, Bhadrak, Kendrapara and Jajpur more than 15% of rice was severely affected by cyclone. A post-analysis ground truth was conducted which showed the accuracy of the rice flooded area mapping was 89%.

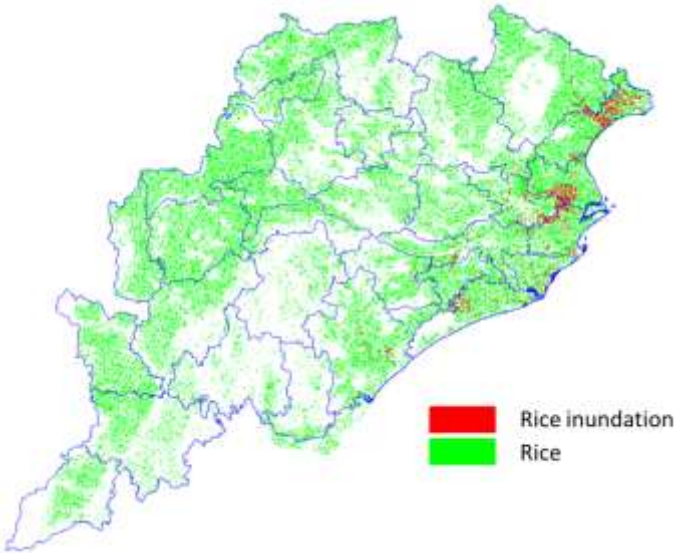


Figure 8. Flooded rice area mapping in Odisha state of Indian post Phailin cyclone

3.3 Rabi Season Crop Alert

As mentioned earlier the drought assessment is generally carried out during the Kharif (Rainy) season. Though majority of crop growing area in Rabi (winter) season is irrigated, it is essential to monitor the agricultural condition to identify any alert situation for necessary intervention measures. The crop condition was assessed using MODIS Vegetation Indices 16-Day Composite data and MODIS Land Surface Temperature 8-day Composite data. Vegetation Condition Index of NDVI and NDWI and Temperature Condition Index (Kogan, 1995) were derived using past ten years' satellite data. Based on the above three mentioned parameters, and by using a logical modeling approach, the district were divided into normal, watch and alert. This exercise was carried out for 8 states namely Bihar, Haryana, Punjab, Rajasthan, Uttar Pradesh, Madhya Pradesh and West Bengal (Figure 9). The analysis showed that the crop situation was normal during the Rabi season of 2013-14.

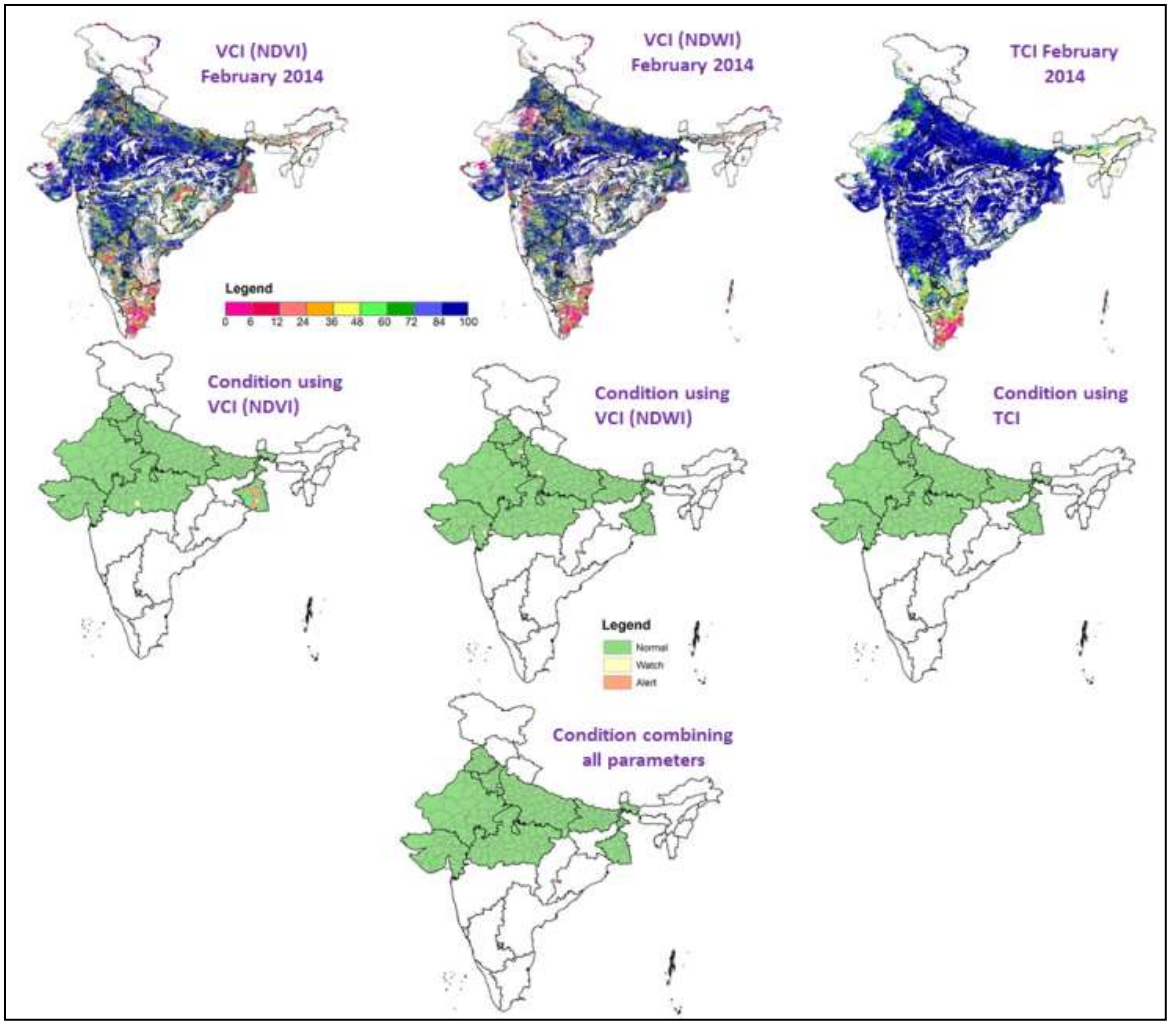


Figure 9. Rabi season crop alert assessment for 8 states of India during February, 2014

4. Conclusions

The examples presented in this article have showed how remote sensing data from various sources, in combination with other ancillary data, have been successfully used for operational assessment of agriculture in the country. However, there is a need to further extend the area of activity. Following future developments are envisaged in this field. i) Taking up more crops and covering more number of states, ii) Developing spectral yield models for all crops, iii) assessing biotic and abiotic disaster impact on agriculture, iv) Monitoring the impact of agricultural development programmes of the country, v) Horticultural assessment, and vii) Agricultural resources management.

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